

Tulio HONORIO DE FARIA

Research Engineer, CEA Paris-Saclay

PERSONAL INFORMATION

Researcher unique identifier(s): **ORCID:** 0000-0003-0281-4502 **Research ID:** S-5867-2017
Google scholar: <https://scholar.google.fr/citations?user=8kHPd58AAAAJ>
Date of birth: **June 20th 1988** Nationality: Brazilian and French

EDUCATION

03/2023: Habilitation (to direct research)
Université Paris-Saclay, France
10/2012-10/2015: PhD, Civil Engineering and Mechanics (laude with honors)
École Normale Supérieure de Cachan, France
PhD supervisor: Benoît Bary; Co-supervisor: Farid Benboudjema
10/2011-09/2012: M.Sc., Materials Science for Sustainable Construction
École des Ponts Paristech, France
Research project supervisor: Bruno Huet (Lafarge)
02/2006-12/2010: BA, Civil Engineering, Federal University of Goiás, Brazil

CURRENT POSITION

From 12/2023: Research Engineer, Concrete and Clay Lab., **CEA Paris-Saclay**, France

PREVIOUS POSITIONS

09/2018-11/2023: Associate Professor, Department of Civil and Environmental Engineering,
École Normale Supérieure Paris-Saclay (ENSPS), France
10/2017-08/2018: Postdoc/Teaching Assistant, MSME Lab., **Université Paris-Est Créteil (UPEC)**
Project: Physical properties of hydroxyapatite from molecular simulations
10/2015-09/2017: Postdoctoral Researcher, **Navier Laboratory**, Ecole des Ponts Paristech (ENPC)
Project: Thermal expansion of adsorbing microporous media and application to clay desiccation
10/2012-10/2015: PhD candidate, Concrete and Clay Lab., **CEA Saclay**:
Project: Modelling and numerical simulation of the mechanical behaviour and cracking of concrete at early age
Project funded by ONDRAF/NIRAS (Belgium)

RESEARCH FUNDING		<i>Status: PI = Principal (Lead) Investigator; CI= Co-Investigator</i>		
	Funding	Value	Title	Status
2019/ 2025	ANR JCJC	284 k€	THEDESCO: Thermal Deformations of Concrete: predictive modeling for tailored design and adaptation https://anr.fr/Projet-ANR-19-CE22-0004 (CI: F. Benboudjema, C. Giry)	PI
2024/ 2026	ANR/ CEA	65 k€	Mastering the expansion in alkali-silica reaction (CI: S. Poyet, T. Charpentier, B. Bary)	PI
2020	DIM RESPORE (Paris region)	60 k€	Capillary Flow Porometer for Multidisciplinary Applications (Co-PI: E. Vennat ; CI: F. Benboudjema, A. Bourdot, J.M. Fleureau)	Co-PI
2023/ 2027	ANR Chaires Industrielles	1000k€	HEAT COFFEE: HEalth evAluaTion of COncrete Fundation For ElEctricity transport https://anr.fr/Projet-ANR-22-CHIN-0007 (PI: F. Benboudjema)	CI
2022/ 2025	CNRS 80 Prime	115 k€	ESOPÉ: Damage by salts: origins of the crystallization pressure (PI: L. Brochard (ENPC) ; Others CI: L. Mercury (ISTO); F. Osselin (ISTO); M. Vandamme (ENPC); J-M. Pereira (ENPC))	CI
2021/ 2024	IDEX LaSIPS	115 k€	HYDRA: Development of a Broad-band Dielectric Approach for the Non-destructive Evaluation of Hydration of	CI

SUPERVISION

Postdoc ((internal funding; *national grants*): 4 (2 concluded)

Ongoing

4. **Fatima Masara** (2026-) “*Structure and elastic properties of (alumino)silicates of magnesium hydrates (M-(A)-S-H)*”, Supervision: T. Honorio, T. Charpentier (CEA), CEA funding

Concluded

3. **Elvys Dias Reis** (2025-2026) “*Mechanisms of expansion in alkali-silice reaction bottom-up*”, Supervision: T. Honorio (CEA), F. Benboudjema, A. Bourdot (ENS Paris-Saclay), **ANR HEAT COFFEE**
2. **Justin Kinda** (2023-2024) “*Experimental evaluation of thermal-sorption coupled deformations in cement-based materials*”, Supervision: T. Honorio, F. Benboudjema (LMPS), **ANR THEDESCO**
1. **Gang Huang** (2023-2024) “*Engineering model for the temperature- and RH- dependence of thermal expansion and heat capacity of concrete*”, Supervision: T. Honorio, F. Benboudjema (LMPS), **ANR THEDESCO**

PhD thesis (internal funding; industrial partners; *national grants*; *thesis director*): 10 (6 defended)

Ongoing

10. **Abderrahmane Lahmar** (2025-) “*Mesosopic simulations and development of simplified models for the mechanical behaviour of irradiated concrete*”. Supervision: **B. Bary (dir.)**, T. Honorio (CEA). CEA funding
9. **Hazem Ferchichi** (2024-) “*Study of the impact of irradiation in cement-based materials: contribution of atomistic methods*”. Supervision: **S. Poyet (dir.)**, P. Fossati, T. Honorio (CEA). CEA funding
8. **Shuai Cheng** (2023-) “*Study at the mesoscale of the delayed behaviour of concrete at 150°C*”. Supervision: **B. Bary (dir.)**, S. Poyet, T. Honorio (CEA); F. Hild (LMPS); J. Sanahuja, J.-L. Adia (EDF R&D)
7. **Syrine Razki** (2023-) “*Study Multiscale characterization of degradation by accelerated aging of concrete foundations*”. Supervision: **F. Benboudjema (dir.)**, A. Bourdot (LMPS); T. Honorio (CEA); F. Hafid (RTE), **ANR HEAT COFFEE**

Defended

6. **Bilal Mahmoud Hawchar** (2022-2025) “*Study of crystallization pressure by molecular simulation and microfluidics experiments*”. Supervision **L. Brochard (dir.)**, M. Vandamme, J.-M. Pereira (ENPC); L. Mercury, F. Osselin (ISTO); T. Honorio (LMPS); **CNRS 80 Prime**
5. **Sofiane Ait-Hamadouche** (2021-2024) “*Development of a Broad-band Dielectric Approach for the Non-destructive Evaluation of Hydration of Cement-Based Materials*”, Supervision **E. Vourc'h (dir.)**, F. Daout (SATIE); F. Benboudjema, T. Honorio (LMPS), **IDEX LaSIPS**
4. **Fatima Masara** (2021-2024) “*Multiscale modelling of the thermo-hygro-mechanical response of concrete informed by the molecular scale*”, Supervision F. Benboudjema, **T. Honorio** (LMPS), **ANR THEDESCO**
3. **Takwa Sayari** (2019-2023) “*Kinetics of drying in concrete: application to achieving water content equilibrium in fire behavior tests.*”, Supervision: **F. Benboudjema (dir.)**, T. Honorio (LMPS); F. Robert and S. Mohaine (**CERIB**)
□ Takwa Sayari received, in May 2022, the special prize sponsored by the National Federation of Public Works
2. **Mihaja Razafimbelo** (2019-2022) “*Optimization of the reinforcement in massive zones in reinforced concrete under complex loadings*”. Supervision: **F. Gatuingt (dir.)**, T. Honorio (LMPS); M. Bottoni and G. Herve-Secourgeon (**EDF R&D**)
1. **Pascale Saba** (2018-2022) “*Mechanics of cement-based composites reinforced by glass fibres grid*”, Supervision: **F. Benboudjema (dir.)**, T. Honorio (LMPS); X. Brajer (**Saint Gobain Research**)

Master project

11. **Syrine Razki** (Mar-2023 to Jun-2023) “*Predicting expansion by external sulphate attack using thermodynamic modelling*” (Co-supervision Alexandra Bourdot)
10. **Bilal Mahmoud Hawchar** (Mar-2022 to Jun-2022) “*Origins of crystallization pressure by molecular simulations*” (Co-supervision Laurent Brochard (ENPC))
9. **Loric Renault** (Apr-2021 to Jun-2021) “*Optimizing the packing density of aggregates in concrete using simulations and machine learning*”
8. **Dikra Afkir** (Mar-2021 to Jul-2021) “*Topology optimization of the reinforcement in cement-based materials*” (Co-supervision Fabrice Gatuingt)
7. **Mohamad Khoder** (Mar-2021 to Jun-2021) “*Experimental characterization and micromechanical modelling of the properties of concrete made with seashell aggregates*” (Co-supervision Alexandra Bourdot)
6. **Sofiane Ait Hamadouche** (Mar-2021 to Jun-2021) “*Nanomechanics of C-S-H and ASR gels: a study by coarse-grained simulations*”
5. **Fatima Masara** (Apr-2020 to Jul-2020) “*Thermal pressurization and effective interactions in C-S-H under sorption*” (Co-supervision Farid Benboudjema)
4. **Estefanía Nuñez Del Prado** (Mar-2020 to Jun-2020) “*Molecular modelling of the hysteresis under sorption and RH-dependent elastic constants of C-S-H*” (Co-supervision Farid Benboudjema)
3. **Nejd Hammami** (Mar-2020 to Jun-2020) “*Influence of aggregate seashell aggregate shape on concrete properties: experimental characterization and micromechanics modelling*” (Co-supervision Alexandra Bourdot, Dalal Badreddine)
2. **Percy Guerra** (Mar-2019 to Jun-2019) “*Dimensional stability of ettringite: molecular dynamics study of the RH- and temperature dependencies*” (Co-supervision Alexandra Bourdot)
1. **Reine Fares** (Mar-2015 to Aug-2015) “*Numerical estimation of the multiscale thermo-viscoelastic properties of concrete at early-age*” (Co-supervision Benoît Bary)

Undergraduate Research Projects

14. **Marwane Triffa** (2022) “*Portlandite under irradiation: mechanisms of H_2 diffusion*”
13. **Bilal Mahmoud Hawchar** (2021) “*C-A-S-H structure and properties*”
12. **Marc Vacher** (2020) “*Cement-based composites with seashells fillers*” (Co-supervision Alexandra Bourdot and Maroua Maroufi)
11. **Sofiane Ait-Hamadouche** (2020) “*Tailoring the composition of cement paste for a target elastic response: using machine learning for the exploration of mix design space*”
10. **Wassi Hussain** (2020) “*Characterization of the hydration and modelling of phases present in pastes with seashell*” (Co-supervision Alexandra Bourdot)
9. **Ornella M. Chemgne Tamouya** (2019) “*Molecular modelling of Alkali-Silica Reaction products*”
8. **Estefanía Nuñez Del Prado** (2019) “*Molecular modelling of the elastic constants of C-S-H*” (Co-supervision Farid Benboudjema)
7. **Sirine Al-Dandachli** (2019) “*Energy storage in ettringite: a study based on molecular simulations*” (Co-supervision Alexandra Bourdot)
6. **Thibault Ronzaud** (2019) “*Valorisation of shell as concrete aggregates: experimental and micromechanics study*”
5. **Rodrigo Yanez** (2019) “*Effects of drying in concrete structures layouts obtained from topology optimization*” (Co-supervision Farid Benboudjema)
4. **Percy Guerra** (2018) “*Identification of ettringite behaviour: physical properties estimated by in silico experiments*”
3. **Samy Ben-Hamoudi** (2018) “*Hydrodynamics in C-S-H nanopores: a molecular dynamics study*”
2. **Martin Pegéot** (2018) “*Topology optimization of reinforcement in rendering mortars*”, (Co-supervision Farid Benboudjema)
1. **Quentin Laurent** (2018) “*Simulation of the evolution of capillary porosity of cement pastes at early age*”

Visiting researchers

4. **Walter Batista Bonfim**, doctoral level (Fev-2026 to Jul-2026) “*Impedance of C-S-H and multiscale models of electrical properties for low carbon cement systems*”. Visiting Researcher from Federal University of Goiás (Brazil).
3. **Naohiko Saeiki**, doctoral level (Sep-2024 to Jul-2025) “*Carbonation of C-S-H*”. Visiting Researcher from University of Tokyo (Japan).
2. **Waleska Barbosa**, post-doc level (Apr-2020 to Dec-2020) “*Molecular modelling of substitutions in and symmetry transitions in tricalcium silicates*”. Visiting Researcher from Federal University of Paraná (Brazil)
1. **Saied Babaei**, doctoral level (Apr-2020 to Jun-2020) “*Micromechanics of drying and thermal shrinkage in cement-based materials*”. Visiting Researcher from SCK-University of Antwerp (Belgium). (Co-supervision F. Benboudjema).

TEACHING ACTIVITIES

2018-2023	ENSPS	Multiscale modelling (M.Sc. level); Complex media; Numerical methods; Soil Mechanics; Energy distribution; Discontinuous media; Transport in porous media
2017-2018	UPEC	Fluid and Solid mechanics; Thermal science; Numerical analysis and Calculus, Numerical simulation of structures; Dynamics and vibrations; General Mechanics
2016-2017	ENPC	Molecular simulations in materials science
2013-2015	ESTP/ESITC	Construction materials

EDUCATIONAL AND ADMINISTRATIVE RESPONSIBILITIES

2022-2023	Head of the group “Life-Cycle of materials, buildings and infrastructures for the reduction of environmental impact of constructions” of the OMEIR team at the LMPS
2021-2023	Head of Master 2 "Materials and Structures" and of the common core with Master 2 “Building envelopes and sustainable construction”
2019-2020	Responsible for the organisation of conferences for M1 and M2 FESUP level at ENS Paris-Saclay
2019	Referent “skills-based approach” for the master of the Department of Civil Engineering and Environment

MEMBERSHIP OF SCIENTIFIC AND TECHNICAL COMMITTEES

From Mar-2025	Chair, RILEM Technical Committee ASM: Atomistic Simulations for Cement-Based Materials: Recommendations and link to Experiments
From Oct-2024	Member, RILEM Technical Committee SDM: Scientific Metadata Management of Construction materials
From Oct-2022	Member, RILEM Technical Committee MMS – Moisture transport and retention in porous media
From May-2022	Member, RILEM Technical Committee DCS – Data-driven concrete science
From Aug-2019	Member, RILEM Technical Committee CCS – Early age and long-term crack width analysis in RC Structures
Jan-2023-Nov-2023	Team Leader, LMPS, DIM MaTerRE (<i>Matériaux avancés éco-responsables – Area of Major Interest of Île-de-France region</i>) https://www.dim-materre.fr/annuaire/tulio-honorio-de-faria/
Feb-2019 to Dec-2021	Team Leader, LMT, DIM RESPORE (Network of Excellence in Porous Solids – Area of Major Interest of Île-de-France region)
Jan-2015 to Mar-2019	Member, RILEM Technical Committee CMS - Thermal cracking of massive concrete structure
Oct-2017 to Oct-2018	Member, International Association ConCreep (IA-CONCREEP)
Jan-2016 to Nov-2018	Member, COST Action TU 1404 <i>Towards the next generation of standards for service-life of cement-based materials and structures</i>
Oct-2014 to Oct-2017	Member, NANOCEM : Partner Project on <i>Multiscale estimation of the thermo-viscoelastic properties of cementitious materials at early age</i>

REVIEWING ACTIVITIES

Grant review (5)

ANR, French National Research Agency,
AAP Graduate School SIS (Sciences de l'Ingénierie et des Systèmes),
ACS Petroleum Research Fund (ACS PRF) grants,

France (3)
France (1)
USA (1)

International journals (150 reviews of 103 manuscripts)

Verified at: <https://www.webofscience.com/wos/author/record/535317>

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|--|---|
| 1. Ambiente Construído (2) | 23. Journal of Energy Storage (2) |
| 2. ACS Applied Nano Materials (1) | 24. Journal of Engineering Mechanics (5) |
| 3. ACS Earth and Space Chemistry (1) | 25. Journal of Geotechnical and Geoenvironmental Engineering (1) |
| 4. ACS Nano (2) | 26. Journal of Materials in Civil Engineering (5) |
| 5. ACS Omega (1) | 27. Journal of Mechanics and Physics of Solids (2) |
| 6. Advanced Theory and Simulations (2) | 28. Journal of Molecular Modelling (1) |
| 7. AIChE Journal (1) | 29. Journal of Physical Chemistry (2) |
| 8. American Mineralogist (1) | 30. Journal of Rock Mechanics and Geotechnical Engineering (1) |
| 9. Applied Clay Science (5) | 31. Journal of Theoretical, Computational and Applied Mechanics (1) |
| 10. Applied Sciences (8) | 32. Langmuir (3) |
| 11. Buildings (2) | 33. Materials (5) |
| 12. Cement and Concrete Composites (6) | 34. Materials & Design (2) |
| 13. Cement and Concrete Research (38) | 35. Materials Research (1) |
| 14. Ceramics International (1) | 36. Materials and Structures (7) |
| 15. Composite Structures (1) | 37. Mechanics of Time-Dependent Materials (2) |
| 16. Construction and Building Materials (15) | 38. Minerals (2) |
| 17. Fractal and Fractional (2) | 39. RILEM Technical Letters (2) |
| 18. International Journal for Numerical and Analytical Methods in Geomechanics (3) | 40. SN Applied Sciences (1) |
| 19. Journal of Advanced Concrete Technology (7) | 41. Structural Engineering and Mechanics (2) |
| 20. Journal of African Earth Sciences (1) | 42. Sustainability (1) |
| 21. Journal of the American Ceramic Society (2) | |
| 22. Journal of Cleaner Production (1) | |

PhD DEFENSE COMMITTEE (13)

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| 2025 | Tarek Ihaddadene (Nantes Université, France) - "Multiscale modelling of ionic diffusion in cement-based materials: contribution of the atomistic approach by molecular dynamics" – defense committee as referee |
| 2025 | Ouail Reguieg (Université de Toulouse, France) - "Molecular Modeling of the behaviour and nanostructural properties of cementitious C-S-H under thermo-hydro-mechanical loadings" – defense committee as referee |
| 2025 | Félicien Pauly (Sorbonne Université, France) - "Coarse grained simulations of suspensions of flexible clay layers" – defense committee as referee |
| 2025 | Jingyi Leng (Ecole Nationale des Ponts et Chaussées, Institut Polytechnique de Paris, Université Gustave Eiffel, CNRS) - "Modeling of sorption-induced deformations of porous materials relevant for energy and civil engineering applications" – defense committee as referee |
| 2025 | Karim Hakim (Aix-Marseille Université, France) - "Nanocomposites carbon-cement for the reduction of carbon footprint of concrete: a numerical study" – defense committee as referee |
| 2024 | Chakib El Faqir (Université de Grenoble, France) - "Investigation of cement carbonation using simultaneous neutron and X-rays tomography coupled with modelling" – defense committee as referee |
| 2024 | Jack Atallah (Nantes Université, France) - "Multiscale modelling of transport properties for the durability evaluation of reinforced concrete in marine environments" – defense committee as referee |

- 2024 **Rémy Hoarau Belkhiri** (Université Paris-Saclay, France) - "Impact of very early-age irradiation on the chemo-mechanical behavior of cementitious matrices in the medium term"– defense committee as invited person
- 2024 **Tiago Campos Neto** (Universidade Federal de Goiás, Brazil) - "Dynamics of carbonation in concrete structures: investigation with temporal series and artificial neural networks"– defense committee as examiner
- 2024 **Priscilla Bastos** (Universidade Federal de Pernambuco, Brazil) - "Development and evaluation of methods to follow reactivity of supplementary cementitious materials"– defense committee as examiner
- 2023 **Thibaut Herin** (Université Paris-Saclay, France) - "Mechanisms of radiolysis in cement hydrates and the consequences of the H₂ formation in irradiated materials"– defense committee as invited person
- 2023 **Sela Hoeun** (INSA Rennes / Université de Nantes, France) - "Influence of adhesion between the main phases in cement system on the mechanical behaviour of hydrated cement paste"– defense committee as reviewer
- 2023 **Xabier Mendez Aretxabaleta** (Universidad del País Vasco, Spain) - "Atomistic simulation on the nucleation of C-S-H"– defense committee as examiner
- 2024 **Jon López Zorrilla** (Universidad del País Vasco, Spain) - "Development and application of transfer learning-enhanced machine learning atomic potentials"– external reviewer (report authorizing defense, reviewer not present during defense)

PhD FOLLOW-UP COMMITTEE

- 2025 **Hind Krim** (ENS Paris-Saclay) – "Prediction of properties of cement-based materials for the structural risk analysis of foundation"
- 2023-24 **Aurore Lechevallier** (Université Paris-Saclay) – "Early age hydration of iron and aluminum containing binder: LDH formation and stability"
- 2023-24 **Tarek Ihaddadène** (Université de Nantes) - "Study of transport properties in concrete : contribution of the atomistic approach by molecular dynamics "
- 2022-24 **Rémy Hoarau Belkhiri** (CEA, Université Paris-Saclay) - "Irradiation of cement-based materials at early-age"
- 2021-23 **Thibaut Herin** (CEA, Université Paris-Saclay) - "Mechanisms of radiolysis in cement hydrates and the consequences of the H₂ formation in irradiated materials"
- 2018 **Justin Kinda** (ENS Paris-Saclay, EDF) - "Impact of drying rate on delayed strain behavior of cement-based materials - experimental and numerical study"

AWARDS AND HONORS

- 2024 Outstanding Researcher from Goiás Recognized Abroad - FAPEG (Brazil)
- 2021 Beneficiary of the PEDR (premium for doctoral supervision and research)
- 2015 PhD laude with honours
- 2011 Lafarge scholarship for master at *École des Ponts Paristech*, France
- 2011 International student scholarship at *École Normale Supérieure de Cachan* with scholarship – *declined*.
- 2011 Scholarship « Erasmus Advanced master on structural analysis of historical constructions » with scholarship – *declined*.
- 2009 BRAFITEC exchange scholarship Brazil-France

INVITED TALKS (12) AND ADVANCED SCHOOLS (4)

- Nov-2025 *Sorption and sorption-induced deformation in cement systems. Keynote* – II International Conference of Atomistic Simulation of Cementitious Materials (ICASCM) - Nanjing, China.
- Jun-2025 *Molecular modelling as enabling technology for decarbonization of the cement industry. Keynote* – II Brazilian Symposium of Cement Science - São Paulo, Brazil.

- Jan-2025 *Water in Concrete Nanopores: From Durability to Creep. Invited seminar* LGCgE – Université de Lille, France
- Dec-2024 *Matériaux cimentaires. Invited seminar* in the workshop Saint-Gobain « **Les atomes, pour quoi faire ?** », La Défense, France
- Jun-2023 *Deciphering nanoscale processes to understand and design for durability. Invited talk (key note)* at V RILEM International Seminar on performance and Durability of Concrete Structures (DURAR 2023: <https://www.durar2023.com/>), Goiânia, Brazil.
- Jun-2023 *Getting interlayer and surface species in C-S-H right to better understand durability of concrete. Invited talk* at workshop USERS-CNRS IRN on Urban Physics, Cairo, Egypt (online participation)
- May-2023 *Analytical approaches to homogenization micro-to-meso – Advanced school of ETN SUBLime (European Training network – Innovative Training Network Sustainable Building Lime Applications via Circular Economy and Biomimetic Approaches)*, TU Delft, The Netherlands.
- Mar-2022 *Multiphysics behaviour of porous materials from the molecular scale. Invited talk* at the Seminar of Quartz Lab of SupMéca, Saint-Ouen, France.
- Mar-2022 *The molecular scale in cement-based materials investigations: contribution to understanding, prediction, and design for durability. Invited talk* at the Workshop of the R&D FURNAS-ANEEL (UFG) “Service Life”, Federal University of Goiás, Goiânia, Brazil.
- Mar-2022 *Multiscale, Multiphysics, Multitechnique modelling applied to cement-based materials. Advanced school* at for the Doctoral program of Civil engineering at the Federal University of Goiás, Goiânia, Brazil.
- Aug-2020 *The perspective of multi-scale modelling in the research on cement-based materials: contribution for the development of predictive models and non-destructive evaluation. Invited talk* at the Workshop of the R&D FURNAS-ANEEL 0394-1504/2015 (UFG part) – Main results, discussion and perspectives in the field of durability of concrete. Online meeting organized by the Federal University of Goiás, Goiânia, Brazil.
- Mar-2020 *Molecular simulations of nanoporous materials: unveiling the physical origins of the thermo-hygro-mechanical response – Advanced school of ETN ATHOR (European Training network – Innovative Training Network Advanced THERmomechanical multiscale mOdelling of Refractory linings)*. A recorded video party was made available in Moodle ELEANOR (operated by RWTH Aachen as part of the ATHOR project) for use by all students associated with universities related to the ATHOR project, and, the wider refractories community via the moodle, for example, but not restricted to, the Federation for International Refractory Research and Education (FIRE).
- Fev-2020 *How molecular simulations may explain multiphysics phenomena in concrete and clays?, Invited talk* at the The University of Queensland, Brisbane, Australia (Funded by BOOSTER project of ENSPS)
- Nov-2019 *Numerical Nano-Twins and Materiomics : elucidating the origins of porous materials behaviour from material « genome », Invited talk* at the Seminar “Club of affiliates of the LMT”, Cachan, France
- May-2019 *Multiscale methods – Advanced school of CSMA Juniors (French Association Calcul des Structures et Modélisation)*, Ile de Porquerolles, France
- May-2019 *Water in cement-based materials, Invited talk* at the weekly seminar at the Thomas Young Centre, London, United Kingdom
- May-2015 *Multiscale Modelling of the thermo-viscoelastic properties of cement-based materials and structures at early-age, Invited talk* at the International Workshop of Young Doctors in Gemechanics (W(H)YDOC), Champs-sur-Marne, France

CONFERENCE AND MINISIMPOSIUM ORGANISATION AND ACTIVITIES

- 2023 Organization of workshop on creep of materials at ENPC
- 2023 *Member of scientific Committee of the Synercrete 23 - The International RILEM Conference on Synergising expertise towards sustainability and robustness of cement-based materials and concrete structures*, Milos Island, Greece
<https://synercrete.com/>
- 2020/21 *Member of scientific Committee of the International RILEM conference on early-age and long-term cracking in RC structures.* <https://crc2021.org>
- 2020 Organization of mini-symposium **MS436 Multiscale mechanics and physics of civil engineering porous materials** on at the 14th World Congress in Computational Mechanics and ECCOMAS Congress. Co-organization with L. Brochard (ENPC), J. Sanahuja (EDF R&D) and K. Ioannidou (CNRS).

PUBLICATIONS

HDR: Honorio de Faria, T. (2023) *Multi-Scale Multi-Physics Multi-Technique modeling for establishing composition-(micro)structure-property correlations in cement-based material*. Université Paris-Saclay <<https://hal-centralesupelec.archives-ouvertes.fr/tel-04051504/>> (manuscript in English)

Ph.D. THESIS: Honorio de Faria, T. (2015) *Modelling concrete behaviour at early-age: multiscale analysis and simulation of a massive disposal structure*. École Normale Supérieure de Cachan. <<https://tel.archives-ouvertes.fr/tel-01241585>> (manuscript in English)

PUBLICATIONS IN INTERNATIONAL PEER-REVIEWED JOURNALS (corresponding author)

· Post-Doc[#]; PhD candidate^{*}; Master Student^{**}; Undergraduate student^{***}

In Preparation

- J75.** Razki, S.^{*}; Bourdot, A.; Benboudjema, F.; Honorio, T. *Transport properties of mortar affected by ASR : A multiscale approach combining thermodynamic modeling and micromechanics*
- J74.** Honorio, T.; Rubens Silve, M.; Julich, L.^{*}; Kirchheim, A.P. *pyPCE: An Open-Source Python Tool for Automated All-Atom Generation of Polycarboxylate Ether Superplasticizers*
- J73.** Masara, F.^{*}; Benboudjema, F.; Honorio, T. *Role of disjoining pressure in drying shrinkage*
- J72.** Masara, F.^{*}; Benboudjema, F.; Honorio, T. *Nanoporosity thermal pressurization in layered adsorbing materials*
- J71.** Masara, F.[#]; Charpentier, T.; Honorio, T. *M-S-H atomic structures*
- J70.** Dias Reis, E.[#]; Razki, S.^{*}; Charpentier, T.; Benboudjema, F.; Honorio, T. *Coexistence of ASR products and C-(N/K)-S-H*
- J69.** Tararushkin, E.V.; Smirnov, G.S.; Honorio, T. *Atomistic modelling of AFm phases: interlayer structure, anion exchange, and the electrostatic origin of solubility*

Under Review

- J68.** Honorio, T.; Saeki, N.^{*}; Barthélemy, J.F.; Maruyama, I.; *Thermodynamically and structurally consistent atomic models of C-S-H*

- J67.** [Cheng, S.*](#); [Poyet, S.](#); **Honorio, T.**; Hild, F.; Bary, B. *Apparent thermal contraction of cement pastes up to 175°C: Experimental characterization, multiscale model, repercussions at the concrete scale*

Under Revision

- J66.** **Honorio, T.** *Power-law creep in C-S-H emerging from subdiffusive dynamics*

2026

- J65.** [Mahmoud Hawchar, B.*](#); **Honorio, T.**; Vandamme, M.; Osselin, F.; Pereira, J.-M.; Mercury, L.; [Brochard, L.](#) (2026) *Fluorescence-based detection of apparent crystallization pressure in microfluidic channels*. JOURNAL OF CRYSTAL GROWTH
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PEER-REVIEWED CONFERENCE PROCEEDINGS

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- C36.** **Ait Hamadouche, S.***; **Honorio, T.**; Bore, T.; Benboudjema, F.; Daout, F.; Vourc'h, E. Multiscale modeling of the dielectric response of C-S-H. SynerCrete 2023, Milos, Greece, June, 2023
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